Spin Parity and Period Obstructions

Research note draft

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Abstract

This note addresses the two parts of Chen–Yu's Problem 3.2 for quadratic differentials. It gives a section-count interpretation of spin parity on the canonical double cover, and it gives a flat orientation-cover invariant distinguishing the regular and irregular exceptional components, with the component identification supplied by Chen–Möller's theorem.

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1 Introduction

Problem 3.2 in the survey of Chen and Yu [2] asks for two refinements in the topology of strata of quadratic differentials. The first is to recover the spin parity of the canonical double cover by computing the dimension of the corresponding theta-characteristic. The second is to find a flat-geometric invariant distinguishing the regular and irregular connected components of the exceptional strata in genera three and four.

The exceptional strata under discussion are

$$Q(9,-1), \quad Q(6,3,-1), \quad Q(3,3,3,-1)$$

in genus three, and

$$Q(12), \quad Q(9,3), \quad Q(6,6), \quad Q(6,3,3), \quad Q(3,3,3,3)$$

in genus four. Chen and Möller proved that the genus-three exceptional strata have two components, called regular and irregular, distinguished by a Brill–Noether condition involving one

third of the zero divisor [1, Theorem 6.1]. In genus four, [1, Theorem 7.1] gives the analogous distinction on the two non-hyperelliptic regular/irregular components. The strata $Q(6,6)$, $Q(6,3,3)$, and $Q(3,3,3,3)$ also have hyperelliptic components; statements below about regular and irregular genus-four components are made on the non-hyperelliptic part.

The second part of this note uses Chen–Möller's component theorem as an input. The new point is to realize their Brill–Noether condition as an invariant on the canonical orientation double cover and as an anti-invariant Mittag–Leffler period obstruction. Thus the result is a conservative solution of the flat-geometric part of Problem 3.2, not an independent reproof of the Chen–Möller component theorem.

2 Spin parity from theta-characteristic sections

Let $(X,q) \in Q(m_1,\dots,m_n)$ be a primitive quadratic differential with at most simple poles, so $m_i \geq -1$. Assume that

$$m_i \not\equiv 2 \pmod 4$$

for all i. Let

$$\pi : \widetilde{X} \to X$$

be the canonical double cover and write

$$\pi^* q = \widetilde{\omega}^2.$$

The cover is ramified exactly over odd-order singularities, including simple poles. Since q is primitive, the cover is connected. Under the congruence assumption all singularities of $\widetilde{\omega}$ have even order, so

$$\widetilde{L} = \mathcal{O}_{\widetilde{X}}\left(\frac{1}{2}\operatorname{div}(\widetilde{\omega})\right)$$

is a theta-characteristic. We use the standard convention that the parity of a theta-characteristic L is $h^0(L) \bmod 2$. Lanneau's constancy result and parity formula are stated via Arf-invariant methods [3]; the following proof gives the requested h^0-interpretation.

Theorem 2.1 (Spin parity formula)**.** *With the notation above,*

$$h^0(\widetilde{X},\widetilde{L}) \equiv 1 - g + \sum_i \left\lfloor \frac{m_i+1}{4} \right\rfloor \pmod 2.$$

Proof. Write

$$\operatorname{div}(q) = \sum_i m_i z_i.$$

Set

$$A = \sum_i \left\lfloor \frac{m_i}{2} \right\rfloor z_i, \qquad B = \sum_{m_i \text{ odd}} z_i.$$

Then

$$\operatorname{div}(q) = 2A + B \sim 2K_X.$$

Let

$$M = K_X(-A).$$

Then

$$M^{\otimes 2} \cong \mathcal{O}_X(B),$$

and the canonical double cover satisfies

$$\pi_* \mathcal{O}_{\widetilde{X}} \cong \mathcal{O}_X \oplus M^{-1}.$$

Define

$$D_+ = \sum_i \left\lfloor \frac{m_i + 1}{4} \right\rfloor z_i.$$

We prove

$$\pi_* \widetilde{L} \cong \mathcal{O}_X(D_+) \oplus K_X(-D_+). \tag{1}$$

The divisor $\frac{1}{2}\operatorname{div}(\widetilde{\omega})$ is invariant under the deck involution τ, so $\widetilde{L}$ carries its natural τ-linearization. We decompose $\pi_* \widetilde{L}$ into invariant and anti-invariant summands.

The calculation is local. If $m = 4r$, the cover is unbranched and locally $\widetilde{\omega} = \pm z^{2r} dz$ on each sheet. Thus $\widetilde{L}$ has coefficient r above the point. The invariant part gives $\mathcal{O}_X(rz)$. Multiplication of an anti-invariant local section by $\widetilde{\omega}$ descends to a differential vanishing to order r, giving $K_X(-rz)$. This is the desired contribution because $\lfloor (4r+1)/4 \rfloor = r$.

If $m = 4r + 1$, put $z = w^2$. Then

$$\pi^* q = (w^2)^{4r+1} (2w\, dw)^2 = 4w^{8r+4} dw^2,$$

so $\operatorname{ord}(\widetilde{\omega}) = 4r+2$, and $\widetilde{L}$ has coefficient $2r+1$ at the ramification point. An invariant local section has only even Laurent terms, and the largest possible pole is $w^{-2r} = z^{-r}$, yielding $\mathcal{O}_X(rz)$. The largest anti-invariant pole is $w^{-(2r+1)}$. Multiplying by $\widetilde{\omega} = 2w^{4r+2} dw$ gives $2w^{2r+1} dw = z^r dz$, so the anti-invariant summand gives $K_X(-rz)$. This matches $D_+ = rz$.

If $m = 4r + 3$, again $z = w^2$ and

$$\pi^* q = (w^2)^{4r+3} (2w\, dw)^2 = 4w^{8r+8} dw^2.$$

Thus $\operatorname{ord}(\widetilde{\omega}) = 4r + 4$, and $\widetilde{L}$ has coefficient $2r + 2$. The invariant part permits even Laurent poles up to order $2r + 2$, hence gives $\mathcal{O}_X((r + 1)z)$. The highest allowed Laurent pole is even and therefore does not occur in the anti-invariant summand. The largest anti-invariant pole is $w^{-(2r+1)}$, and

$$w^{-(2r+1)} \cdot 2w^{4r+4} dw = 2w^{2r+3} dw = z^{r+1} dz.$$

Thus the anti-invariant summand is $K_X(-(r + 1)z)$, matching $D_+ = (r + 1)z$.

If $m = -1$, locally $q = z^{-1} dz^2$. With $z = w^2$, one has $\pi^* q = 4 dw^2$, so $\widetilde{\omega}$ is regular and nonvanishing. Hence D_+ receives no contribution. The invariant part gives $\mathcal{O}_X$, and an anti-invariant holomorphic function is $wh(w^2)$. Multiplying by dw gives $\frac{1}{2} h(z) dz$, an arbitrary holomorphic differential on the base. Thus the anti-invariant summand is K_X. Finally, if $m = 0$, the cover is unbranched and $\widetilde{\omega}$ is nonvanishing; the local splitting is $\mathcal{O}_X \oplus K_X$.

These local computations prove (1). Therefore

$$h^0(\widetilde{X}, \widetilde{L}) = h^0(X, \mathcal{O}_X(D_+)) + h^0(X, K_X(-D_+)).$$

By Riemann–Roch,

$$h^0(X, \mathcal{O}_X(D_+)) - h^0(X, K_X(-D_+)) = \deg D_+ + 1 - g.$$

Modulo 2, sum and difference agree. Hence

$$h^0(\widetilde{X},\widetilde{L}) \equiv \deg D_+ + 1 - g = 1 - g + \sum_i \left\lfloor \frac{m_i+1}{4} \right\rfloor \pmod 2.$$

□

3 The orientation-cover invariant

Let (X,q) be in one of the genus-three exceptional strata, or in one of the non-hyperelliptic regular/irregular genus-four exceptional strata listed above. Let $Z(q)$ be the zero divisor of q, and define

$$D_q = \frac{1}{3}Z(q).$$

In genus three, the simple pole is not included in D_q. Let

$$\pi : \widetilde{X} \to X, \qquad \pi^* q = \omega^2$$

be the canonical orientation double cover, with deck involution τ. Then $\tau^*\omega = -\omega$. Define

$$E_q = \frac{1}{2}\pi^* D_q, \qquad I_{\mathrm{oc}}(q) = h^0(\widetilde{X}, \mathcal{O}_{\widetilde{X}}(E_q))^-.$$

Lemma 3.1. *The divisor E_q is integral.*

Proof. If q has zero order $3a$ at z, then D_q has coefficient a at z. If a is even, then the orientation cover is unbranched over z, and $\frac{1}{2}\pi^*(az)$ has coefficient $a/2$ at each of the two points above z. If a is odd, the cover is ramified and $\pi^*(az) = 2a\widetilde{z}$, so $\frac{1}{2}\pi^*(az) = a\widetilde{z}$. Hence all coefficients are integral. □

Theorem 3.2 (Orientation-cover invariant)**.** *There is a natural isomorphism*

$$H^0(X, K_X(-D_q)) \cong H^0(\widetilde{X}, \mathcal{O}_{\widetilde{X}}(E_q))^-.$$

Consequently,

$$I_{\mathrm{oc}}(q) = h^0(X, K_X(-D_q)).$$

Proof. Define

$$\Phi : H^0(X, K_X(-D_q)) \to H^0(\widetilde{X}, \mathcal{O}(E_q))^-$$

by

$$\Phi(\eta) = \frac{\pi^*\eta}{\omega}.$$

It is anti-invariant because $\pi^*\eta$ is invariant and ω is anti-invariant. We check the pole bound. If q has zero order $3a$ at z and a is even, the cover is unbranched and $\mathrm{ord}(\omega) = 3a/2$ above z. Since η vanishes to order at least a, the quotient has order at least $-a/2$, exactly the allowed pole order given by E_q. If a is odd, write $z = w^2$. Then

$$\pi^* q = 4w^{6a+2}dw^2, \qquad \omega = 2w^{3a+1}dw.$$

If $\eta = z^s dz$ with $s \geq a$, then $\pi^*\eta = 2w^{2s+1}dw$. Therefore

$$\mathrm{ord}(\pi^*\eta/\omega) \geq (2a+1) - (3a+1) = -a,$$

again exactly the pole bound prescribed by E_q. At a simple pole, $q = z^{-1}dz^2$, and with $z = w^2$ one has $\pi^* q = 4dw^2$. Thus ω is regular and nonvanishing. The pole is not in D_q, so E_q has coefficient zero there, and the quotient is holomorphic.

Conversely, let $f \in H^0(\widetilde{X}, \mathcal{O}(E_q))^-$. Then $f\omega$ is τ-invariant and hence descends to a meromorphic differential η on X. The preceding calculations in reverse show that η is holomorphic and vanishes along D_q. Indeed, in the unbranched case the order of $f\omega$ is at least $-a/2 + 3a/2 = a$. In the ramified case the order is at least $-a + (3a+1) = 2a+1$, which descends to a differential $z^s dz$ with $s \geq a$. At a simple pole, since E_q has coefficient zero and f is anti-invariant, locally $f(w) = c_1 w + c_3 w^3 + \cdots$. Thus

$$f\omega = 2(c_1 w + c_3 w^3 + \cdots)dw = (c_1 + c_3 z + \cdots)dz,$$

which descends holomorphically. Hence $\eta \in H^0(X, K_X(-D_q))$, and the two constructions are inverse. □

4 Anti-invariant period obstruction

The invariant in Theorem 3.2 can be formulated as a finite-dimensional Mittag–Leffler obstruction on the orientation cover. Let $\mathcal{P}_q$ be the vector space of principal parts of meromorphic functions with poles bounded by E_q along $\mathrm{Supp}(E_q)$. It is naturally a representation of τ. Let

$$\delta_q : \mathcal{P}_q \to H^1(\widetilde{X}, \mathcal{O}_{\widetilde{X}})$$

be the Mittag–Leffler coboundary map, and let

$$\delta_q^- : \mathcal{P}_q^- \to H^1(\widetilde{X}, \mathcal{O}_{\widetilde{X}})^-$$

be its anti-invariant part. Then

$$\ker \delta_q^- \cong H^0(\widetilde{X}, \mathcal{O}_{\widetilde{X}}(E_q))^-. \tag{2}$$

Theorem 4.1 (Anti-invariant obstruction criterion)**.** *In the range of Theorem 3.2,*

$$\ker \delta_q^- \cong H^0(X, K_X(-D_q)).$$

In particular,

$$I_{\mathrm{oc}}(q) > 0 \quad \Longleftrightarrow \quad \delta_q^- \text{ is not injective.}$$

Proof. By (2) and Theorem 3.2,

$$\ker \delta_q^- \cong H^0(\widetilde{X}, \mathcal{O}(E_q))^- \cong H^0(X, K_X(-D_q)).$$

This proves the theorem. □

There is an equivalent period formulation. Let $P \in \mathcal{P}_q^-$. Locally P is a principal part of a meromorphic function, hence dP has zero residues at all points of $\mathrm{Supp}(E_q)$. Choose an anti-invariant meromorphic differential β_P on $\widetilde{X}$ whose principal part is dP. This is possible by the usual Mittag–Leffler theorem for differentials of the second kind, because the prescribed residues are zero. The ambiguity in such a choice is an anti-invariant holomorphic differential.

Fix an anti-invariant symplectic basis

$$\gamma_1, \dots, \gamma_{2g^-}$$

of $H_1(\widetilde{X}, \mathbb{Z})^-$. The periods of β_P, modulo the anti-invariant holomorphic ambiguity, define a linear obstruction map

$$\mathrm{Per}_q^- : \mathcal{P}_q^- \to \mathbb{C}^{2g^-} / \mathrm{Per}(H^0(\widetilde{X}, K_{\widetilde{X}})^-).$$

The condition $\mathrm{Per}_q^-(P) = 0$ means that one can choose β_P with all anti-invariant periods equal to zero. In that case integration gives a global anti-invariant meromorphic function with principal part P, so $P \in \ker \delta_q^-$. Conversely, if P is the principal part of a global anti-invariant meromorphic function, then its differential has vanishing periods. Therefore

$$\ker \mathrm{Per}_q^- = \ker \delta_q^-.$$

This is the flat period-obstruction form of I_{oc}.

5 Residual cubic and the component distinction

Let $Q(\mu)$ be one of the exceptional quadratic strata in genus 3 or 4. Thus

$$\mu = (3a_1, \dots, 3a_n, -1), \qquad \sum_i a_i = 3,$$

in genus 3, and

$$\mu = (3a_1, \dots, 3a_n), \qquad \sum_i a_i = 4,$$

in genus 4. In genus 4 we work on the non-hyperelliptic regular/irregular components. For $q \in Q(\mu)$, put

$$D_q = \frac{1}{3} Z(q).$$

Let P_q be the simple-pole divisor in genus 3, and put $P_q = 0$ in genus 4. Then

$$\mathrm{div}(q) = 3D_q - P_q.$$

Definition 5.1 (Residual cubic and positive loci)**.** *We say that q admits a residual cubic Abelianization if there are Abelian differentials $\eta, \theta \in H^0(X, K_X)$ and an effective divisor $R \in X^{(g-2)}$ such that*

$$\mathrm{div}(\eta) = D_q + R, \qquad \mathrm{div}(\theta) = 3R + P_q,$$

and

$$q = \frac{\eta^3}{\theta}.$$

We denote the residual cubic Abelianization locus by

$$\mathcal{R}_\mu^{\mathrm{cub}} = \{q : q \text{ admits a residual cubic Abelianization}\},$$

and the positive orientation-cover locus by

$$\mathcal{E}_\mu = \{q : I_{\mathrm{oc}}(q) > 0\}.$$

Lemma 5.2 (Residual cubic equivalence)**.** *For the exceptional strata above,*

$$h^0(X, K_X(-D_q)) > 0$$

if and only if q admits a residual cubic Abelianization. Equivalently,

$$\mathcal{E}_\mu = \mathcal{R}_\mu^{\mathrm{cub}}.$$

Proof. Suppose first that $0 \neq \eta \in H^0(X, K_X(-D_q))$. Then

$$\operatorname{div}(\eta) = D_q + R$$

for an effective divisor R of degree $g-2$. Define

$$\theta = \frac{\eta^3}{q}.$$

Since $\operatorname{div}(q) = 3D_q - P_q$, we have

$$\operatorname{div}(\theta) = 3(D_q + R) - (3D_q - P_q) = 3R + P_q.$$

Thus θ is holomorphic and $q = \eta^3/\theta$.

Conversely, if $q = \eta^3/\theta$ with θ holomorphic, then

$$\operatorname{div}(q) = 3\operatorname{div}(\eta) - \operatorname{div}(\theta).$$

At every zero of q belonging to D_q, the effectivity of $\operatorname{div}(\theta)$ forces η to vanish at least to the order prescribed by D_q. Hence

$$0 \neq \eta \in H^0(X, K_X(-D_q)).$$

The equality $\mathcal{E}_\mu = \mathcal{R}_\mu^{\mathrm{cub}}$ follows from Theorem 3.2. □

Theorem 5.3 (Component distinction)**.** *For each exceptional genus-three stratum and for each non-hyperelliptic regular/irregular genus-four exceptional stratum,*

$$\mathcal{E}_\mu = \mathcal{R}_\mu^{\mathrm{cub}} = \{q : h^0(X, K_X(-D_q)) > 0\}.$$

Equivalently, since $\deg D_q = g$*,*

$$q \in \mathcal{E}_\mu \quad \Longleftrightarrow \quad h^0(X, \mathcal{O}_X(D_q)) \geq 2.$$

By Chen–Möller's theorem [1, Theorems 6.1 and 7.1], this condition is exactly the traditional irregular component condition in genus three and on the non-hyperelliptic regular/irregular part in genus four. Its complement is the corresponding regular component.

Proof. The equality

$$\mathcal{E}_\mu = \{q : h^0(X, K_X(-D_q)) > 0\}$$

is Theorem 3.2. The equality with $\mathcal{R}_\mu^{\mathrm{cub}}$ is Lemma 5.2. For the listed exceptional signatures, $\deg D_q = g$, so Riemann–Roch gives

$$h^0(X, \mathcal{O}_X(D_q)) - h^0(X, K_X(-D_q)) = 1.$$

Thus $h^0(X, K_X(-D_q)) > 0$ is equivalent to $h^0(X, \mathcal{O}_X(D_q)) \geq 2$. Chen–Möller prove that this Brill–Noether condition distinguishes the exceptional regular and irregular components: in genus three this is [1, Theorem 6.1], and in genus four, on the non-hyperelliptic regular/irregular components, this is [1, Theorem 7.1]. Combining their component theorem with the orientation-cover and period-obstruction identifications above proves the claim. □

Remark 5.4. *The component identification in Theorem 5.3 uses Chen–Möller's Brill–Noether component theorem as an input. The point of the preceding sections is that the same condition is naturally visible on the orientation double cover and through anti-invariant flat periods.*

6 Conclusion

Theorem 2.1 gives the requested section-count interpretation of spin parity. Theorems 3.2, 4.1, and 5.3 give the flat-geometric invariant for the exceptional regular/irregular distinction: the invariant is I_{oc}, equivalently the kernel of the anti-invariant period-obstruction map on the orientation cover. The final identification with the traditional component names relies on Chen–Möller's component theorem.

References

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